



Secular coefficients mini-course

Talk 3

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We are interested in understanding the coefficients of the characteristic polynomial of a random unitary matrix (known as secular coefficients); these are analogous to character sums.

The characteristic polynomial $\det(I - uU_N)$ tends in distribution to the simpler series $\exp(-\sum_{i=1}^{\infty} N_i u^i / \sqrt{i})$.

If χ_m is a random Dirichlet character modulo m , then $L(s, \chi_m)$ tends to $\sum \alpha(n)/n^s$ where α is the Steinhaus multiplicative function.

So for now, we are studying the simpler c_n , which are the coefficients of the random model $\exp(-\sum_{i=1}^{\infty} N_i u^i / \sqrt{i})$.

Similarly, we study the simpler partial sums of the Steinhaus function

$$\frac{1}{\sqrt{x}} \sum_{n \leq x} \alpha(n)$$

instead of character sums.

Number theorists have considered random multiplicative functions motivated not only by the study of Dirichlet characters or the analogy with matrices. The initial motivation was the study of the Möbius function:

$$\mu(p_1 \cdots p_k) = (-1)^k$$

if p_i are distinct primes.

The Riemann Hypothesis is equivalent to the statement

$$\sum_{n \leq x} \mu(n) = O_\varepsilon(x^{\frac{1}{2} + \varepsilon}).$$

In fact, many deep problems in number theory concern square-root cancellation, e.g. the Generalized Riemann Hypothesis is the same as $\sum_{n \leq x} \mu(n) \chi(n) = O_\varepsilon(x^{1/2 + \varepsilon})$.

Random walk

Naive probabilistic model for the partial sums $\sum_{n \leq x} \mu(n)$ (or variants): a random walk $\sum_{i=1}^n X_i$ where X_i are iid random variables.

Well understood:

- 1 CLT (Central Limit Theorem): $\frac{1}{\sqrt{n}} \sum_{i=1}^n X_i$ tends in distribution to a Gaussian.
- 2 Moments tend to Gaussian moments.
- 3 We have the LIL (Law of Iterated Logarithm):
$$\limsup_{n \rightarrow \infty} \frac{\sum_{i=1}^n X_i}{\sqrt{2n \log \log n}} = 1 \text{ almost surely.}$$
- 4 Gaussian tails: $\mathbb{P}(|\sum_{i=1}^n X_i| > \sqrt{n\lambda}) \ll e^{-c\lambda^2}$.

Random multiplicative functions

The conjectural square-root cancellation for multiplicative sums cannot be modeled by a random walk: in a random walk, increments are independent.

However, for a multiplicative function f , the value at n is determined by previous values $(f(i))_{i < n}$ (unless n is a prime, which is rare), e.g. $\mu(6) = \mu(2)\mu(3)$. No independence.

Wintner (1944) was the first to explicitly consider modeling deterministic multiplicative functions by random multiplicative functions.

While the study of $\sum_{n \leq x} \alpha(n)$ is more accessible than the study of deterministic sums, our understanding of it is still quite poor compared to random walks.

Almost sure behavior (I)

Wintner (1944): $\sum_{n \leq x} \alpha(n) = O(x^{1/2+\varepsilon})$ almost surely,
 $\sum_{n \leq x} \alpha(n) \neq O(x^{1/2-\varepsilon})$ almost surely. Random RH.

Proof: through the random Dirichlet series

$$F(s) = \sum_{n=1}^{\infty} \frac{\alpha(n)}{n^s}.$$

Wintner shows $F(s)$ converges in $\Re s > 1/2$ almost surely;
almost surely cannot be extended to $\Re s \geq 1/2$.

Relationship: let $A(x) := \sum_{n \leq x} \alpha(n)$. Then

$$F(s) = s \int_1^{\infty} A(x) x^{-s-1} dx.$$

Almost sure behavior (II)

Harper (2020): Almost surely

$|\sum_{n \leq x} \alpha(n)| \geq x^{1/2} (\log \log x)^{1/4 - \varepsilon}$ along a diverging sequence of values of x . Answered conjectures of Halász and Erdős.

Caich (2024): $|\sum_{n \leq x} \alpha(n)| \leq x^{1/2} (\log \log x)^{3/4 + \varepsilon}$ almost surely.

There is no conjecture for the optimal function $f(x)$ such that $\limsup_{x \rightarrow \infty} |\sum_{n \leq x} \alpha(n)|/f(x) = 1$ almost surely.

Moments

Even moments were studied by various authors, but do not reveal much as they diverge quickly:

$$\mathbb{E} \left| \frac{1}{\sqrt{x}} \sum_{n \leq x} \alpha(n) \right|^{2k} \sim C_k (\log x)^{(k-1)^2}.$$

Low moments studied by Harper (2017). Notably,

$$\mathbb{E} \left| \frac{1}{\sqrt{x}} \sum_{n \leq x} \alpha(n) \right| \asymp (\log \log x)^{-\frac{1}{4}}.$$

Harper asked about limiting distribution once normalized correctly.

CLT

Theorem 1 (G.-Wong, 2025)

We have

$$\frac{\sum_{n \leq x} \alpha(n)}{\sqrt{x}(\log \log x)^{-\frac{1}{4}}} \xrightarrow[x \rightarrow \infty]{d} \sqrt{V_\infty} G$$

where V_∞ is independent of $G \sim \mathcal{N}_{\mathbb{C}}(0, 1)$, and will be discussed later.

Combined with Harper's moment bounds immediately implies moment asymptotics ('uniform integrability'):

$$\lim_{x \rightarrow \infty} \mathbb{E} \left| \frac{\sum_{n \leq x} \alpha(n)}{\sqrt{x}(\log \log x)^{-\frac{1}{4}}} \right|^{2q} = \mathbb{E}[|V_\infty|^q] \Gamma(q+1)$$

for $q < 1$.

Theorem 1 was preceded by

Theorem 2 (G.-Wong, 2025)

Let $\mathcal{A}$ be a multiplicative set, i.e. $a, b \in \mathcal{A} \implies ab \in \mathcal{A}$, and $ab \in \mathcal{A}$ implies $a, b \in \mathcal{A}$ if $\gcd(a, b) = 1$.

If $\sum_{p \in [1, x] \cap \mathcal{A}} \sim \theta x / \log x$ for $\theta \in (0, 1)$, then

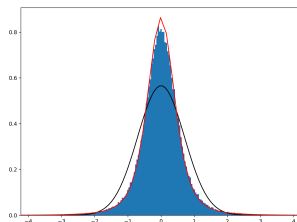
$$\frac{\sum_{n \in [1, x] \cap \mathcal{A}} \alpha(n)}{\sqrt{|[1, x] \cap \mathcal{A}|}} \xrightarrow[x \rightarrow \infty]{d} \sqrt{V_{\infty}^{\mathcal{A}}} G$$

where $V_{\infty}^{\mathcal{A}}$ is independent of $G \sim \mathcal{N}_{\mathbb{C}}(0, 1)$.

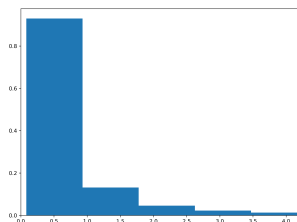
Example: the set $\mathcal{A} = \text{SOS of sums of two squares}$ ($\theta = 1/2$).

Soundararajan–Xu: if one takes $n \in [x, x + h] \cap \text{SOS}$ where $h = h(x)$ is $o(x)$, one obtains a Gaussian limit (McLeish).

Pictures



Real part of sum, plotted against **Gaussian** and **Gaussian mixture**.



V_∞ is heavy-tailed: $\mathbb{P}(V_\infty > t) = t^{-1+o(1)}$ (Wong).

Theorem 2 is motivated by, and structured similarly to, a pair of papers on secular coefficients by Najnudel–Paquette–Simm and Najnudel–Paquette–Simm–Vu.

For every $\beta > 0$ one has a probability measure on $U(N)$ that depends on β , known as $C^\beta E$. If one denotes a matrix sampled from $C^\beta E$ by $U_{N,\beta}$ then

$$\det(I - uU_{N,\beta}) \xrightarrow{d} \exp \left(-\sqrt{\frac{2}{\beta}} \sum_{i=1}^{\infty} N_i \frac{u^i}{\sqrt{i}} \right).$$

N–P–S studied secular coefficients with parameter $\beta > 4$; N–P–S–V did so when $\beta > 2$. The case $\beta = 2$ was resolved by Atherfold–Najnudel; it corresponds to Haar measure.

How is $C\beta E$ related to Theorem 2? As usual, it is instructive to look at finite fields. Suppose α_q is the Steinhaus function on $\mathbb{F}_q[T]$, and $\mathcal{A}$ is a completely multiplicative set in $\mathbb{F}_q[T]$ such that

$$\sum_{P \in \mathcal{A}, \deg P = n} 1 \sim \frac{2}{\beta} \frac{q^n}{n}.$$

Let

$$\mathcal{L}_{\mathcal{A}}(u, \alpha_q) := \sum_{f \in \mathcal{A}} \alpha_q(f) u^{\deg f} = \prod_{P \in \mathcal{A}} (1 - \alpha_q(P) u^{\deg P})^{-1}.$$

Then by applying the CLT one sees that (informally)

$$\log \mathcal{L}_{\mathcal{A}}(u/\sqrt{q}, \alpha_q) \approx \sqrt{\frac{2}{\beta}} \sum_{i=1}^{\infty} N_i \frac{u^i}{\sqrt{i}}.$$

Log-correlated fields

Let $F: \mathbb{R} \rightarrow \mathbb{R}$ be a random function ('field'), such that

$$\mathbb{E}F(x) \equiv 0, \mathbb{E}[F(x)F(y)] \approx \frac{1}{2} \log \frac{1}{|x-y|}.$$

That is, values at two points are correlated, and correlation decays logarithmically (in particular – slowly) as distance grows.

This does not make sense at $x = y$, so often one considers a family of fields with covariance $\frac{1}{2} \log \min\{\frac{1}{|x-y|}, T\}$.

Typically, $F(x)$ is approximately Gaussian at each x .

Ubiquitous in mathematics: probability, number theory, random matrix theory

Examples

Example 1: $t \mapsto \Re \left[\sum_{k=1}^N \frac{N_k}{\sqrt{k}} e^{ikt} \right]$ for iid $N_k \sim \mathcal{N}_{\mathbb{C}}(0, 1)$.

Proof: Covariance at t_1 and t_2 is $\sum_{k \leq N} \frac{1}{k} \Re[N_k e^{ikt_1}] \Re[N_k e^{ikt_2}] = \Re \sum_{k \leq N} \frac{e^{ik(t_1 - t_2)}}{2k} \approx \log \min\{|t_1 - t_2|^{-1}, N\}/2$.

Example 2: $t \mapsto \log |F_T(1/2 + it)|$ for $F_T(s) = \prod_{p \leq T} (1 - \alpha(p)p^{-s})^{-1}$, partial Euler product for α .
In fact, Bourgade (2009): $t \mapsto \log |\zeta(1/2 + iTU + it)|$ is log-correlated, for $U = \text{Unif}([0, 1])$.

Multiplicative Chaos

Let $t \mapsto F(t)$ be a logarithmically correlated field. Multiplicative chaos is a *random measure* defined, roughly, by

$$g \mapsto \int_{\mathbb{R}} g(t) \frac{e^{\gamma F(t)}}{\mathbb{E} e^{\gamma F(t)}} dt.$$

Often defined as a limit (because logarithmically correlated fields are often obtained as limits of sequences of fields).

Multiplicative chaos measures are multifractal, in the sense that $\mathbb{E}[\mu(I)^p] \asymp |I|^{c(p)}$ as $|I| \rightarrow 0$, where $c(p)$ is non-linear.

Multiplicative Chaos (II)

Let

$$\mu_{\theta,T}(dt) := \frac{e^{2\theta \log |F_T(1/2+it)|}}{\mathbb{E} e^{2\theta \log |F_T(1/2+it)|}} dt = \frac{|F_T(1/2+it)|^{2\theta}}{\mathbb{E} |F_T(1/2+it)|^{2\theta}} dt$$

be a random measure on $[0, 1]$.

Saksman and Webb showed $\mu_{\theta,T}(dt)$ tends to a non-trivial random measure $\mu_{\theta}(dt)$ when $\theta < 1$ (multiplicative chaos measure).

They also showed $\sqrt{\log \log T} \mu_{1,T}(dt)$ tends to a non-trivial random measure (critical multiplicative chaos measure).

Multiplicative Chaos (III)

Elementary computation:

$$\mathbb{E} \int_0^1 (\Re \log F_T(1/2 + it) - \theta \log \log T)^2 \mu_{\theta, T}(dt) = O(\log \log T).$$

This means that $\mu_{\theta, T}$ is concentrated on values of t where

$$\log |F_T(1/2 + it)| = \theta \log \log T + O(\sqrt{\log \log T}).$$

However, Arguin–Belius–Harper proved that almost surely,

$$\max_{t \in [0, 1]} \log |F_T(1/2 + it)| = \log \log T - \frac{3 + o(1)}{4} \log \log \log T.$$

When $\theta = 1$, the contribution to the integral comes from a small set.

Definitions of V_∞ and $V_\infty^{\mathcal{A}}$

Recall $F_T(s) = \prod_{p \leq T} (1 - \alpha(p)p^{-s})^{-1}$. For a subset $\mathcal{A}$ of primes, let

$$F_T^{\mathcal{A}}(s) = \prod_{p \leq T, p \in \mathcal{A}} (1 - \alpha(p)p^{-s})^{-1}.$$

Then

$$V_\infty = \lim_{T \rightarrow \infty} \frac{\sqrt{\log \log T}}{2\pi} \int_{\mathbb{R}} \frac{|F_T(1/2 + it)|^2}{\mathbb{E}[|F_T(1/2 + it)|^2]} \frac{dt}{|1/2 + it|^2},$$
$$V_\infty^{\mathcal{A}} = \lim_{T \rightarrow \infty} \frac{1}{2\pi} \int_{\mathbb{R}} \frac{|F_T^{\mathcal{A}}(1/2 + it)|^2}{\mathbb{E}[|F_T^{\mathcal{A}}(1/2 + it)|^2]} \frac{dt}{|1/2 + it|^2}.$$

Roughly, these are $\lim_{T \rightarrow \infty} \int_{[0,1]} \mu_{\theta,T}(dt) = \int_{[0,1]} \mu_\theta(dt)$, the total mass of multiplicative chaos measures.

Why do $V_{\infty}^{\mathcal{A}}$ and V_{∞} appear? (I)

It is known that, in many cases, the Dirichlet series of an arithmetic function governs its behavior, e.g. Perron's formula:

$$\sum_{n \leq x} \alpha(n) = \frac{1}{2\pi i} \int_{(c)} \sum_n \frac{\alpha(n)}{n^s} \frac{x^s}{s} ds.$$

However, using Perron's formula directly does not seem to help in studying the distribution and does not explain the appearance of V_{∞} .

Why do $V_\infty^{\mathcal{A}}$ and V_∞ appear? (II)

Instead, the appearance of V_∞ can be motivated by Plancherel's identity for Dirichlet series (which can be deduced from Perron's): given a function $f: \mathbb{N} \rightarrow \mathbb{C}$, let

$$F_f(s) = \sum_{n=1}^{\infty} \frac{f(n)}{n^s}.$$

Then

$$\int_{\mathbb{R}} \frac{|F_f(c+it)|^2}{|c+it|^2} dt = 2\pi \int_0^\infty \frac{|\sum_{n \leq x} f(n)|^2}{x^{1+2c}} dx$$

for every c where convergence holds.

Some proof ideas of Theorems 1 and 2

Starting point: as in the previous talk, we use the martingale decomposition

$$\sum_{n \leq x} \alpha(n) = \sum_{p \leq x} S_p$$

for

$$S_p = \sum_{n \leq x: P(n)=p} \alpha(n)$$

where $P(n)$ is the largest prime factor of n . The expectation of S_p , when conditioned on $\alpha(q)$ for $q < p$, is 0.

McLeish CLT is only relevant when the limiting distribution is Gaussian, so cannot be applied here. Instead, we turn to the martingale central limit theorem. It was also used by Najnudel–Paquette–Simm when studying the coefficients of $\exp(-\sqrt{2/\beta} \sum_{i \geq 1} N_i u^i / \sqrt{i})$, $\beta > 4$.

Theorem (Martingale CLT)

In the setting of McLeish, $\sum_{i \geq 1} X_{i,n} \xrightarrow[n \rightarrow \infty]{d} \sqrt{V} \mathcal{N}_{\mathbb{C}}(0, 1)$ holds if

- $\sum_{i \geq 1} \mathbb{E}[X_{i,n}^2 \mid \mathcal{F}_{i-1}] \xrightarrow[n \rightarrow \infty]{p} 0$,
- $\sum_{i \geq 1} \mathbb{E}|X_{i,n}|^4 \rightarrow 0$ as $n \rightarrow \infty$,
- and $V_n := \sum_{i=1}^{k_n} \mathbb{E}[|X_{i,n}|^2 \mid \mathcal{F}_{i-1}] \xrightarrow[n \rightarrow \infty]{p} V$.

We apply the theorem to our martingale decomposition. The theorem tells us that Theorem 1 is proved if

$$V_x := \frac{\sqrt{\log \log x}}{x} \sum_{p \leq x} \mathbb{E} \left[|S_p|^2 | \alpha(1), \dots, \alpha(p-1) \right] \xrightarrow[p \rightarrow \infty]{} V_\infty.$$

Rearranging, the left-hand side ('bracket process') is essentially

$$V_x = \sqrt{\log \log x} \sum_{p \leq x} \frac{1}{p} \left| \frac{\sum_{n \leq x/p, P(n) < p} \alpha(n)}{\sqrt{x/p}} \right|^2.$$

A rather horrendous random variable.

One naive approach is to prove $\mathbb{E}|V_x - V_\infty|^2$ tends to 0, by expanding and using orthogonality. Unfortunately, this moment blows up.

However, this strategy works in the context of Theorem 2 for the appropriate difference $V_x^{\mathcal{A}} - V_\infty^{\mathcal{A}}$, but only if $\theta < 1/2$.

In general, the moment $\mathbb{E}|V_\infty^{\mathcal{A}}|^q$ is finite if $q < 1/\theta$.

What if $\theta \in (1/2, 1]$? We approximate

$$V_x = \sqrt{\log \log x} \sum_{p \leq x} \frac{1}{p} \left| \frac{\sum_{n \leq x/p, P(n) < p} \alpha(n)}{\sqrt{x/p}} \right|^2$$

by

$$\approx \sqrt{\log \log x} \int_2^x \left| \sum_{n \leq x/t, P(n) < t} \alpha(n) \right|^2 \frac{dt}{t \log t},$$

and find a way to relate this to $F(s) = \sum_n \alpha(n)/n^s$ using Plancherel.

Two obstacles in applying Plancherel to

$$\sqrt{\log \log x} \int_2^x \left| \sum_{n \leq x/t, P(n) < t} \alpha(n) \right|^2 \frac{dt}{t \log t}.$$

- 1 Plancherel relates partial sums of a function f to the Dirichlet series of f . Here we have *many* f :
 $f_t(n) = \alpha(n) \mathbf{1}_{P(n) < t}$ for each t . To get around this we discard atypical terms from the original sum, and justifying this is a significant part of the proof.
- 2 When applying Plancherel one must choose a c . The random variable V_∞ is defined by working on the critical line so one wants to take $c = 1/2$. However, it turns out the c that arises naturally here is $1/2 + \lambda/\log x$ for some λ ; one has to show that if one tweaks the definition of V_∞ the limit is unchanged ('universality' of multiplicative chaos).

Thank you for listening.